

Aishwarya Ponkshe

Product Manager

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SUMMARY

Engineer turned Product Manager with experience driving product lifecycles across global markets. Leverages deep expertise in thermal engineering, regulatory compliance, and product benchmarking to solve complex system challenges. Recent technical achievements include developing an AI Scarcity Simulation to evaluate LLM psychological profiles and performing global market studies in automotive sector.

WORK EXPERIENCE

Cummins Inc | System Performance Senior Engineer | Columbus, IN Aug 2023 - Jan 2026

Owled engine calibration strategy for high altitude (12000 ft, 60F) and cold ambient (-40F) operation across 3 Cummins CARB/EPA 2027 and LATAM 2025 programs by aligning product performance with regulatory and customer requirements.

- Drove **160k annual fuel savings** for a 100-truck fleet and **3% improved fuel economy** over legacy methodology by conducting customer-usage driven performance analysis.
- Generated **240k annual savings in personnel and facility costs** by defining off-nominal calibration best practices which accelerated test-cell schedule by 1 week per project.
- Built data-driven tools in MATLAB + Excel and generated tuning tables of combustion variables to provide calibration revisions remotely for field tests and customer issue resolution.
- Analyzed **500+ datasets** to diagnose field performance issues and provide interim solutions to customers in USA, France and Peru with an **average turnaround time of 2 days** per request.
- Improved **process reliability by 30%** for engine friction characterization by redesigning cold motoring (-40F - 0F) sequence across 20+ automated tests to prevent low pressure thermal events.
- Managed end-to-end test execution for **20+ projects** within in-house and external test facilities to oversee engine install, DVPR planning, logistics & calibration release and presented calibration content in leadership design reviews.

Ray W. Herrick Labs, Purdue University | Research Assistant | West Lafayette, IN Aug 2021 - Aug 2023

Led a team of 5 researchers to deliver proof of concept models, identify gaps in existing workflows in simulation-based product development, while serving as the primary industry liaison.

- Developed an integrated 1D-3D finite element thermal model of a 6-cylinder diesel engine to predict in-cylinder heat transfer and temperature distribution and **achieved ± 10 °C accuracy versus hardware** thermal mapping.
- Built and delivered curriculum for ME440: Automotive Prime Movers to 70+ undergraduate students.

Cummins Inc | System Simulation Engineer Intern | Remote, USA Jun 2022 - Aug 2022

- Quantified model uncertainty in engine performance predictions by building a DOE-based sensitivity analysis in GT Power models, identifying key component variation drivers in 19 - 95 L engines.
- Scaled operational infrastructure by driving migration of 50+ models from a siloed storage system to Ansys Minerva database which enabled revision control and standardized workflows across business unit.

Cummins Inc | Product Engineering & Management | Pune, India Oct 2019 - Jun 2021

Developed engineering solutions in product design, quality control & systems integration for new product development programs across four business units while building end-to-end knowledge of Cummins's product lifecycle.

- Identified **30% size reduction opportunity** for future residential genset platforms by leading competitive benchmarking and market analysis and determined usage-driven design trade-offs through reverse engineering.
- Delivered the launch of **2000 kVA 'No Cool' genset series** accounting for **20% of India's total manufacturing volume** by owning critical **Go-to-Market milestones** such as service documentation, OEM validation, and transportation feasibility.
- Drove **\$600k annual cost savings** by validating a local supplier for fuel injector subassembly owning end to end change management process from performance verification to teardown sign-off.
- Compiled product requirements documents for CPCB IV+ engine programs by translating customer specifications, emissions criteria, hazard & safety, hardware limitations, and installation guidelines into engineering requirements.
- Aligned critical system interface agreements for BS VI programs across multiple business units, clarifying ownership and integration readiness for key deliverables.

EDUCATION

The Wharton School | Strategic Management Business Certification Online

Purdue University | Master of Science in Mechanical Engineering West Lafayette, IN

Savitribai Phule Pune University | Bachelor of Engineering in Mechanical Engineering Pune, India

SKILLS

Product: Market Research, Benchmarking, Product Roadmap, Requirements Management, Competitive Advantage Frameworks, New Product Introduction, Product Lifecycle Management, FMEA, 7-Step Problem Solving, Root Cause Analysis

System Analysis & Test: GT Power, MATLAB, SIMULINK, SQL, Python, Ansys Chemkin, Calterm (ETAS INCA equivalent), Excel, Power BI, AutoCAD, SOLIDWORKS, PTC Integrity, Design of Experiments